



Connecting a Cloud Storage bucket to Gemini Enterprise

1. When to use this

Use a Cloud Storage data store when you want a **curated, reviewed corpus**: policy PDFs, handbooks, benefits summaries, FAQ documents. You decide exactly what goes in, and the content does not change under you because someone edited a file in their personal drive.

2. Before you start

#	What you need
1	A Google Cloud project with billing enabled
2	A Gemini Enterprise app, or permission to create one
3	A Cloud Storage bucket in the same or a different project
4	Documents in a supported format, listed in section 3
5	Editor or Owner on the project, or the Discovery Engine Admin role

3. Prepare the bucket

Supported unstructured formats: **PDF, HTML, DOC, TXT, PPTX**.

Two limits worth knowing before a large import:

- A single import matches at most **100,000 files**.
- Per-file size limits are documented in *Prepare data for ingesting*. At time of writing the ceiling is 200 MB per file. Verify this before you plan a bulk load, because it changes.

Import is **recursive**. Everything inside subfolders of the path you give is pulled in too, so an over-broad `gs://` path quietly balloons the job. Point at a dedicated folder, not the bucket root, unless you genuinely mean the whole bucket.

Practical advice on file naming: the filename often shows up in citations, so `03-benefits-and-leave-guide.pdf` reads far better to an end user than `doc_final_v2_USE_THIS.pdf`.

```
None
gs://your-bucket-name/
  01-employee-handbook.pdf
  02-new-hire-checklist.pdf
  03-benefits-and-leave-guide.pdf
  04-401k-faq.pdf
  05-benefits-summary.pdf
```



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Search (/) for resources, docs, products, and more

Search

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Cloud Storage

Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Configuration

Bucket details

onboarding-gem-ent-492719

Location: us-central1 (Iowa) | Storage class: Standard | Public access: Not public | Protection: Soft Delete

Objects | Configuration | Permissions | Protection | Lifecycle | Observability | Operations

Folder browser

onboarding-gem-ent-492719

More results

onboarding-gem-ent-492719

Create folder | Upload | Transfer data | Create batch operations | Other services | Learn

Filter by name prefix only | Filter | Filter objects and folders | Show | Live objects only

	Name	Size	Type	Created	Storage class	Last modified	
☐	01-altostrat-employee-handbook.p...	6.8 KB	application/pdf	Aug 20, 2026, 1:25:06 AM	Standard	Aug 20, 2026, 1:27:49 AM	
☐	02-new-hire-day-one-checklist.pdf	8.1 KB	application/pdf	Aug 20, 2026, 1:25:06 AM	Standard	Aug 20, 2026, 1:25:06 AM	
☐	03-benefits-and-leave-guide.pdf	7.6 KB	application/pdf	Aug 20, 2026, 1:25:06 AM	Standard	Aug 20, 2026, 1:25:06 AM	
☐	04-expense-and-travel-policy.pdf	7.8 KB	application/pdf	Aug 20, 2026, 1:25:06 AM	Standard	Aug 20, 2026, 1:25:06 AM	
☐	05-security-and-acceptable-use-po...	7.9 KB	application/pdf	Aug 20, 2026, 1:25:06 AM	Standard	Aug 20, 2026, 1:25:06 AM	

4. Create the data store

Go to the Google Cloud console, open **Gemini Enterprise**, then **Data Stores**, then **Create Data Store**.

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Search

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Gemini Enterprise

Apps

Data stores

Manage users

Usage & Spending

Developer tools metrics

Settings

Create data store

Source

Data

Configuration

Update: Set up connections faster using a single data source card. Select your data source (e.g., Jira) and then choose your connection mode (Ingestion or Federated Search), if applicable, in the next screen.

Select a data source

Choose a data source for your data store

Search by tool name, Agent Registry ID, type, producer, or use case

First-party data sources

First-party Google data sources that you can connect to your instance.

Website Content

Automatically crawl public website content from a list of URL patterns you define.

Add data source

BigQuery

Import data from your BigQuery table.

Add data source

Cloud Storage

Import data from your storage bucket.

Add data source

Third-party data sources

Third-party data sources that you can connect to your instance.

On the source selection page, choose **Cloud Storage**.

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Search

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Configuration

Import data from Cloud Storage

Specify the data you want to import to your data store

What kind of data are you importing?

For more information, see [Prepare data for ingesting](#).

Structured Data Import

Import general structured data files.

☐ Tabular/E-commerce

Import data that is entirely structured, like product catalogs or BigQuery tables. The schema is auto-detected.

☐ JSONL with document IDs

The file must follow the specific root-level schema. You must provide an ID for each document.

☐ CSV (for FAQ data)

The file must contain structured FAQ data.

☐ Media/Catalog Assets

JSONL files referencing media assets or product SKUs (Specialized structured data). [View requirements](#)

Unstructured Data Import (Document Search & RAG)

Import specific data types that require specialized handling, such as unstructured files or media.

☐ Documents with Metadata (RAG)

JSONL files used to link metadata to unstructured documents (PDFs, HTML, etc.). The schema is auto-detected. [View requirements](#)

☒ Documents

Unstructured documents like PDFs, HTML, TXT, DOCX, and PPTX. Supports healthcare documents (PDF, XML, etc.) and other document types.

☐ Documents with Healthcare data

[Preview](#)
Selecting this option applies specific processing and compliance measures for unstructured healthcare data.

5. Choose one-time or periodic ingestion

This is the most consequential choice on the page, and it is easy to click past.



	One-time (GA)	Periodic (Public preview)
Refresh	Manual re-import	Automatic on a schedule
Frequency options	Not applicable	Daily, every 3 days, or every 5 days
First sync	Immediate	About one hour after creation
Later syncs	Not applicable	Roughly every 24, 72, or 120 hours
Manual refresh	Yes	No
Data source access control	Supported	Not supported
Multiple buckets or folders in one store	Yes	One entity data store per connector

Choose **one-time** unless the documents genuinely change on their own. Policy PDFs and handbooks are republished deliberately, not continuously, and one-time ingestion gives you access control plus the ability to merge several folders into a single store.

Choose **periodic** when an upstream process writes into the bucket on a schedule and you accept losing access control.

Synchronization frequency

This action cannot be changed after the data store is created

☒ One time

One time ingestion only

☐ Periodic

Periodic data ingestion

☐ Streaming

Preview

Continuous data update. The default speed is 100 document per second.

6. Select the folder or file

Under **Select a folder or file you want to import**, pick **Folder** or **File**.

Click **Browse** and choose your data, then **Select**. Alternatively type the path directly into the `gs://` field:

None

gs://your-bucket-name/policies

Then choose the kind of data you are importing. For PDFs, handbooks and policy documents this is **Unstructured documents**.

Click **Continue**.

Streaming

Preview

Continuous data update. The default speed is 100 document per second.

Select a folder or a file you want to import

We will recursively import all files for buckets and folders.

Folder

File

gs:// *

onboarding-gem-ent-492719

Browse

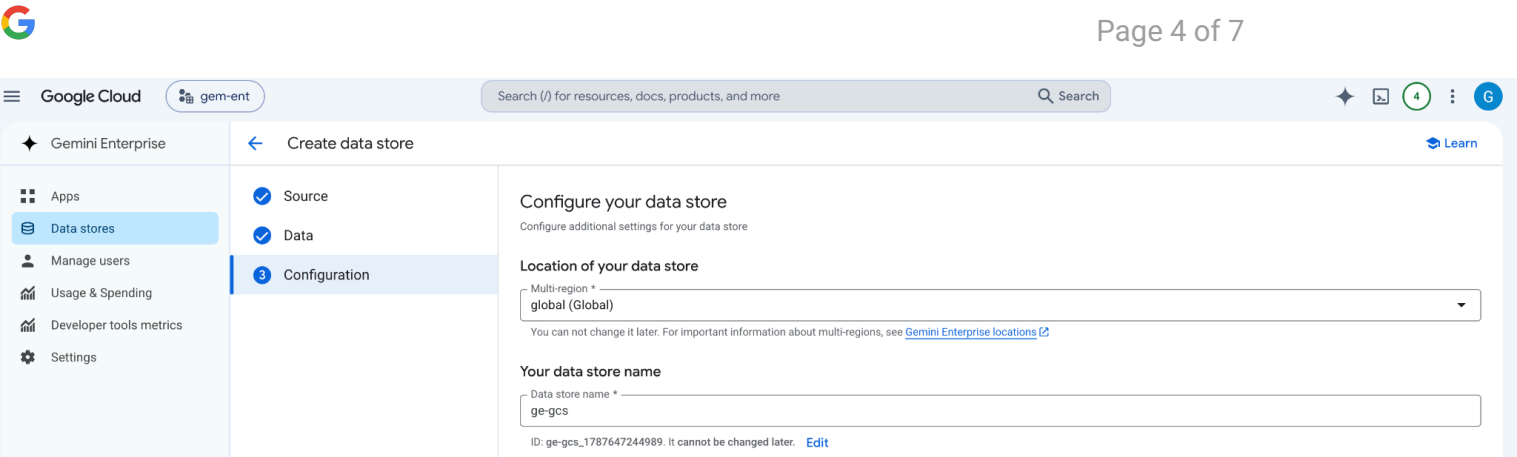
Continue

Cancel

7. Region and name

Pick a region, then give the data store a name.

The region matters more than it looks. Data stores from any region can be used by a tool, but once a tool uses one region, every other data store on that tool must come from the same region. Mixing regions means building a second tool. Pick one region and use it consistently across the project.



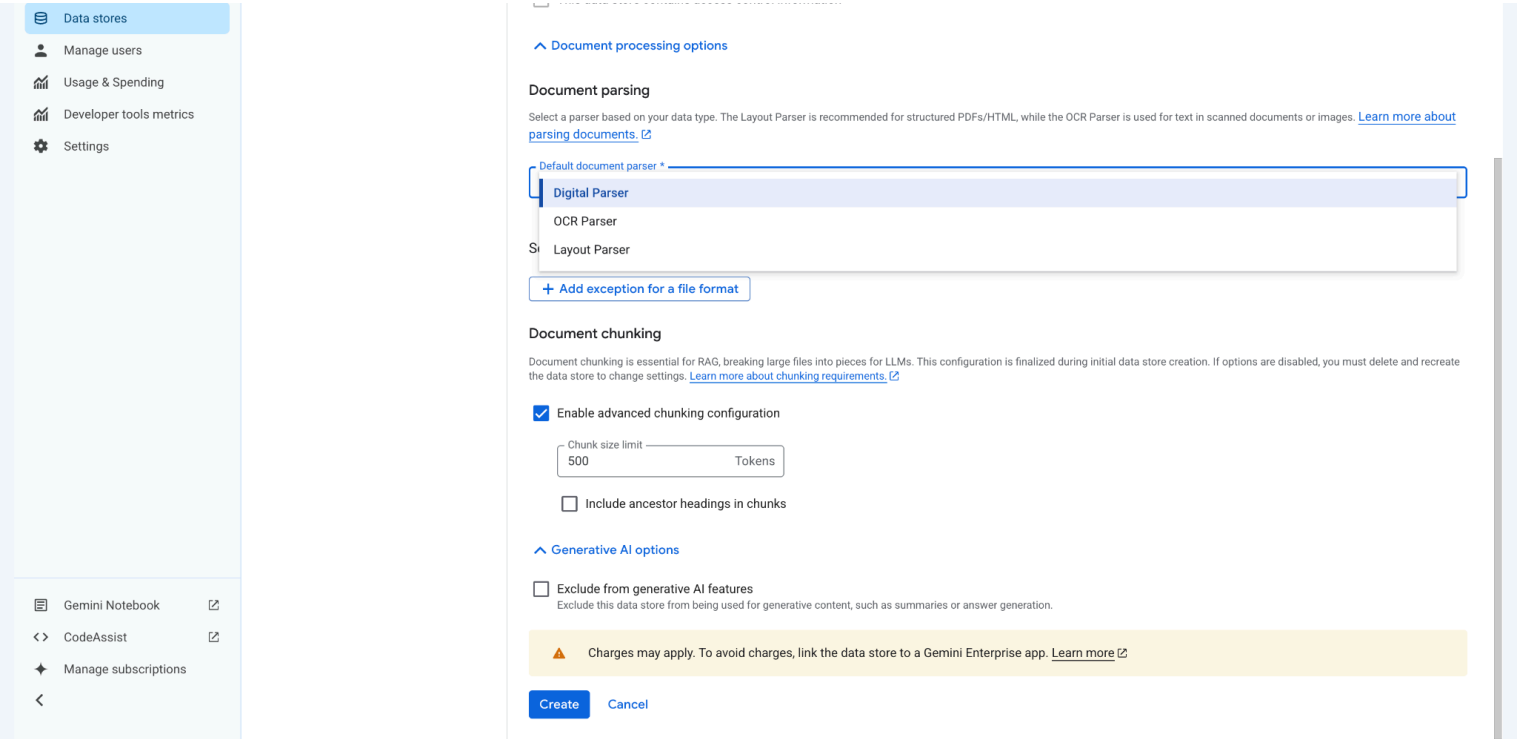
8. Document processing options

For unstructured data you can expand **Document processing options** to control parsing and chunking.

Option	When you want it
Default (digital) parser	Text-based PDFs. Fastest and cheapest
OCR parser	Scanned documents, or text embedded in images
Layout parser	Documents where structure matters: tables, headings, multi-column
Chunking	Turn this on if the data store will be used for retrieval augmented generation

Two warnings the console does not make loud enough:

- The OCR and layout parsers **incur additional costs** and add meaningful processing time.
- **Chunking is chosen at creation time** and is not easily retrofitted. If this data store will ground a Gemini Enterprise assistant, which is the usual reason for creating it, turn chunking on now.



Click **Create**.

9. Verify the import actually finished

Do not skip this. Testing before indexing completes produces answers that look broken and sends you debugging the wrong thing.

Go to **Data Stores**, click your data store, open the **Data** page, then the **Activity** tab. Wait for the status to move from **In progress** to **Import completed**.

For a periodic connector the tab is called **Data ingestion activity**, and you are waiting for **In progress** to become **succeeded**.

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Search (/) for resources, docs, products, and more

Search

Gemini Enterprise

Data

Data stores > onboarding_materials > Data

onboarding_materials

Details

Documents

Events

Activity

Parsing & Processing

Preview

Data store ID

onboarding-materials_1787203915957

Type

Unstructured data

Serving state

Enabled

Region

global

Language

N/A

Connected apps

gemini-enterprise-1

Datastore size

37.31 KiB

Number of documents

5

Last document import

Aug 20, 2026, 1:32:17 AM

View details

Exclude from generative AI features

False

Then open the **Documents** tab and confirm the document count matches what you uploaded. A count lower than expected usually means an unsupported format was silently skipped.

Details

Documents

Events

Activity

Parsing & Processing

Preview

+ Import documents

Purge Data

Filter

Enter property name or value

ID	URI	Index Status	Actions
576b81a7e5d17c16273fbb66b8bce97f	gs://onboarding-gem-ent-492719/03-benefits-and-leave-guide.pdf	Indexed: 8/20/2026, 1:44:39 AM, America/New_York	
64ded6e8c136794b50bb59e25c42aa4	gs://onboarding-gem-ent-492719/04-expense-and-travel-policy.pdf	Indexed: 8/20/2026, 1:44:41 AM, America/New_York	
6ea6ec0c23916d583248eb1cd6be333f	gs://onboarding-gem-ent-492719/01-altostrat-employee-handbook.pdf	Indexed: 8/20/2026, 1:44:44 AM, America/New_York	
96633e101ee7ff4260b5e35afaf63771	gs://onboarding-gem-ent-492719/05-security-and-acceptable-use-policy.pdf	Indexed: 8/20/2026, 1:44:39 AM, America/New_York	
9dcf6c955b05780b436cc218a652cb7b	gs://onboarding-gem-ent-492719/02-new-hire-day-one-checklist.pdf	Indexed: 8/20/2026, 1:44:43 AM, America/New_York	

Expect anywhere from a few minutes for a handful of PDFs to a few hours for a large corpus. Import completing and indexing completing are two different milestones, and the second one is the one that matters. Validate with a real query rather than assuming readiness.

10. Attach the data store to your app

A data store on its own does nothing. Open your Gemini Enterprise app, go to its data sources, and add the data store you just created.

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Search

Gemini Enterprise

Overview

Connected data stores

Actions

Prompt chips

Configurations

Agents

Skills

Security

Integration

Analytics

Apps > gemini-enterprise-1 > Data stores

Connected data stores

+ New data store

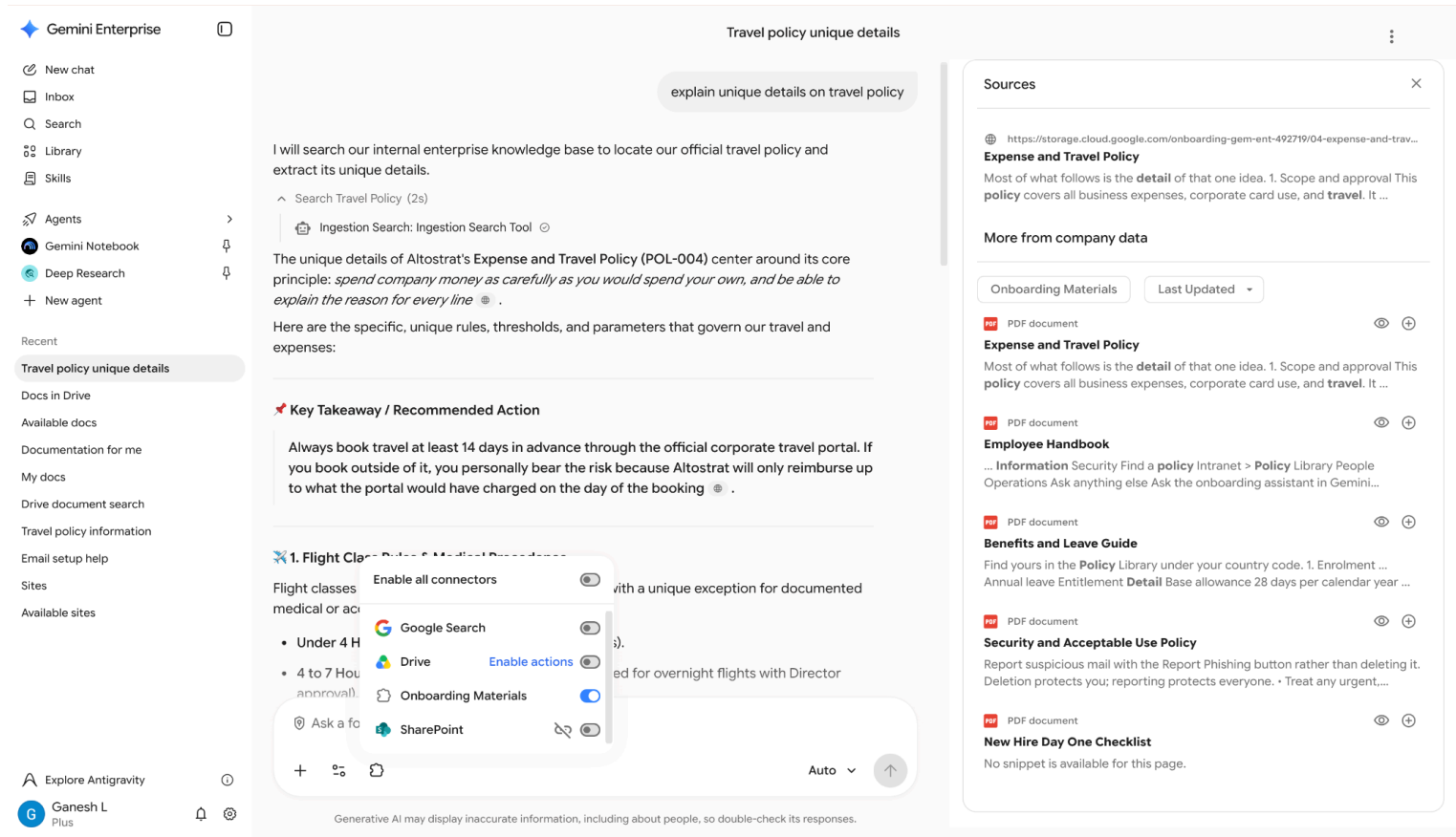
Add existing data stores

Filter

Enter property name or value

Name	Type	Status	Last data sync	Last update	Date created
onboarding_materials	Unstructured data		-		Aug 20, 2026
ge_drive_1	Google Drive	Active		Aug 24, 2026, 3:48:12 PM	Aug 21, 2026
testsharepoint	SharePoint	Active		Aug 24, 2026, 3:48:13 PM	Aug 19, 2026

Then test with a question whose answer you know is in one specific document, and check that the citation points at that document.



11. Cross-project buckets

If the bucket lives in a different project from the data store, grant these two roles to the Discovery Engine service agent, in the project that holds the **data store**:

None

service-PROJECT_NUMBER@gcp-sa-discoveryengine.iam.gserviceaccount.com

- Storage Object Admin (roles/storage.objectAdmin)
- Storage Object Viewer (roles/storage.objectViewer)

Replace PROJECT_NUMBER with the numeric project number, not the project ID.

12. The security point you must not miss

Cloud Storage permissions are not imported with the data.

Whatever IAM you have on the bucket does not follow the documents into the index. Once content is ingested, anyone with sufficient permission on the Gemini Enterprise app can retrieve it through search, regardless of whether they could open the original file in the bucket.

Treat a Cloud Storage data store as **one shared visibility tier**. If some documents must only be visible to some people, either split them into separate data stores with separate apps, or move to a custom connector that carries document-level ACLs.

For HR content this matters. A benefits summary is fine for everyone. A compensation band document is not.

13. Troubleshooting

Symptom	Cause	Fix
Fewer documents indexed than uploaded	Unsupported format silently skipped	Check formats against section 3
Import stuck In progress for hours	Large corpus, or OCR/layout parser on many files	Wait, then check the Activity tab for errors



Symptom	Cause	Fix
Import fails immediately, cross-project bucket	Discovery Engine service agent lacks bucket access	Section 11
Answers are vague or miss obvious content	Chunking was not enabled at creation	Recreate the data store with chunking on
Answers cite the wrong document	Corpus too broad, or near-duplicate documents	Split into smaller focused data stores
Import pulled in far more than expected	Recursive import from too high a path	Point at a specific folder
Periodic connector will not refresh on demand	By design, periodic cannot be manually refreshed	Use one-time ingestion if you need control

14. Cloud Storage connector versus Google Drive connector

Both put documents behind your assistant. They work in fundamentally different ways, and the difference is about **who is allowed to see what**.

	Cloud Storage	Google Drive
How it works	Documents are copied into the search index	Federated. Nothing is copied, results are fetched live
Permissions	Bucket IAM does not carry over. One shared visibility tier	Live Drive sharing, enforced per user at query time
Freshness	One-time, or every 24, 72 or 120 hours	Always current
Prerequisites	A bucket and its contents	Google Workspace customer ID and a configured identity provider
Scope limit	Whatever you put in the folder	Only documents owned by the searching user's domain. Personal gmail.com accounts are not supported
You control content	Yes, explicitly	No, it reflects whatever is in Drive

Choosing between them

Pick **Cloud Storage** when the corpus should be curated and identical for everyone. Published policies, handbooks, benefits documents, FAQs. You decide what is authoritative, and nobody's half-finished draft leaks into an answer.

Pick **Google Drive** when people need to search their own working material and per-user permissions must be respected exactly. Notes, team folders, documents in progress.

For a policy and onboarding assistant, Cloud Storage is almost always the right starting point. The whole value is that every employee gets the same, correct, reviewed answer.